Problem Set 1: The Geometry of Events

Student Handout

Sets, Venn diagrams, De Morgan's laws, and inclusion–exclusion

80 points + 10 optional challenge points

Name: _____ **Due:** _____

Purpose. Sets are the grammar in which probability will be written. This assignment develops fluency in translating among words, symbols, Venn diagrams, and counts. Give reasons for your answers: a diagram may support an argument, but it does not replace one when a proof is requested.
Notation. All complements are taken relative to the stated universe Ω. The symbol $|A|$ denotes the number of elements in a finite set A. The word "or" is inclusive unless explicitly stated otherwise.

I. Sets in symbols and pictures

1. Membership and operations. [10 points] Let

$$\Omega = \{1,2,3,4,5,6,7,8,9,10,11,12\}, \quad A = \{2,4,6,8,10,12\}, \quad B = \{2,3,5,7,11\}.$$

List the elements of each set.

(a) $A \cap B$

(b) $A \cup B$

(c) A^c

(d) $A \setminus B$

(e) $(A \cup B)^c$

2. Negating carefully: De Morgan's laws. [10 points] Let $A, B \subseteq \Omega$.

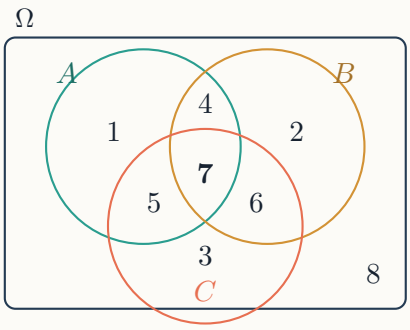
(a) Translate $(A \cup B)^c = A^c \cap B^c$ into an English sentence.

(b) Prove the identity in part (a) by showing that an arbitrary element $x \in \Omega$ belongs to the left side exactly when it belongs to the right.

(c) State the other De Morgan law and prove it in the same way.

3. Reading and shading Venn diagrams. [12 points]

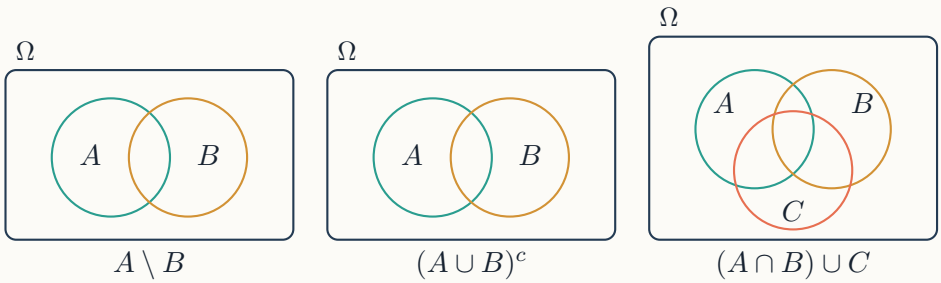
In the numbered diagram below, each number labels one of the eight disjoint regions determined by A, B, and C.



(a) Give the region number(s) belonging to each set:

$$A \cap B \cap C, \qquad A \setminus (B \cup C), \qquad (A \cup B)^c, \qquad (A \cap B) \setminus C.$$

(b) Shade the requested set in each diagram.



II. Counting with overlaps

4. Two-set inclusion–exclusion. [10 points] Among 120 students, 68 study calculus, 54 study programming, and 31 study both. Assume every student is among the 120 being surveyed.

(a) How many study calculus or programming (or both)?

(b) How many study exactly one of the two subjects?

(c) How many study neither subject?

(d) Explain briefly why simply adding $68 + 54$ gives the wrong answer to part (a).

5. A three-set survey. [14 points] At a university with 200 surveyed students, let M, C, E denote the students taking mathematics, computing, and economics. Suppose

$$|M| = 105, \quad |C| = 90, \quad |E| = 75,$$
$$|M \cap C| = 50, \quad |M \cap E| = 40, \quad |C \cap E| = 35,$$
$$|M \cap C \cap E| = 20.$$

The pairwise intersection counts include students taking all three subjects.

(a) Determine the number in each of the eight regions of a three-set Venn diagram.

(b) How many students take at least one of the three subjects?

(c) How many take exactly one? Exactly two? All three? None?

(d) Check that your four answers in part (c) add to 200.

6. Difference distributes over union. [10 points]

(a) Use De Morgan's law and the identity $A \setminus B = A \cap B^c$ to simplify $A \setminus (B \cup C)$.

(b) Prove directly, by following an arbitrary element x, that

$$A \setminus (B \cup C) = (A \setminus B) \cap (A \setminus C).$$

(c) Explain the identity in one ordinary English sentence.

III. From diagrams to arguments

7. Cancellation can fail for sets. [6 points] The equation $a + c = b + c$ implies $a = b$ for numbers. Decide whether each proposed analogue for sets is true. Prove it if it is true; give a concrete counterexample if it is false.

(a) If $A \cup C = B \cup C$, then $A = B$.

(b) If both $A \cup C = B \cup C$ and $A \cap C = B \cap C$, then $A = B$.

8. Why three-set inclusion–exclusion works. [8 points] For finite sets A, B, C, prove

$$|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C|.$$

Do this by choosing one element x and checking its total contribution to the right-hand side when x belongs to exactly zero, one, two, or three of the sets.

Optional challenge

9. Inclusion–exclusion among the integers. [10 optional points] How many integers from 1 through 500 are divisible by at least one of 4, 6, or 9? How many are divisible by none of them? Show how every term in your inclusion–exclusion calculation arises.

Before submitting: check complements against Ω, include overlaps when you write "or," and make every inclusion–exclusion correction visible.